

CHAPTER 4

(15-25 pages..)

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

The **Car Rental Management and Analytics System** was developed to improve the way car rental businesses handled their daily operations. It provided a centralized platform where administrators could manage vehicle information, customer records, reservations, rental transactions, payments, and maintenance schedules more efficiently. In addition to these management functions, the system included an analytics dashboard that presented valuable business information, such as rental trends, vehicle availability, revenue, and other key performance indicators. These features helped business owners monitor their operations and make better decisions based on accurate and organized data.

Before the development of the system, many car rental businesses relied on manual methods of recording information, such as paper documents or spreadsheets. These methods often resulted in delayed transactions, duplicate records, and difficulties in tracking vehicle availability and rental history. To address these challenges, the developed system automated the major processes involved in car rental operations. By storing information in a centralized database, the system reduced manual work, improved the accuracy of records, and made important information easier to access whenever needed.

The system was also designed to be simple and user-friendly so that administrators and staff could easily perform their daily tasks with minimal training. Aside from managing rental transactions, the analytics feature allowed users to view reports and graphical summaries that highlighted rental activities, vehicle utilization, and business performance. These reports provided useful insights that supported planning, monitoring, and decision-making.

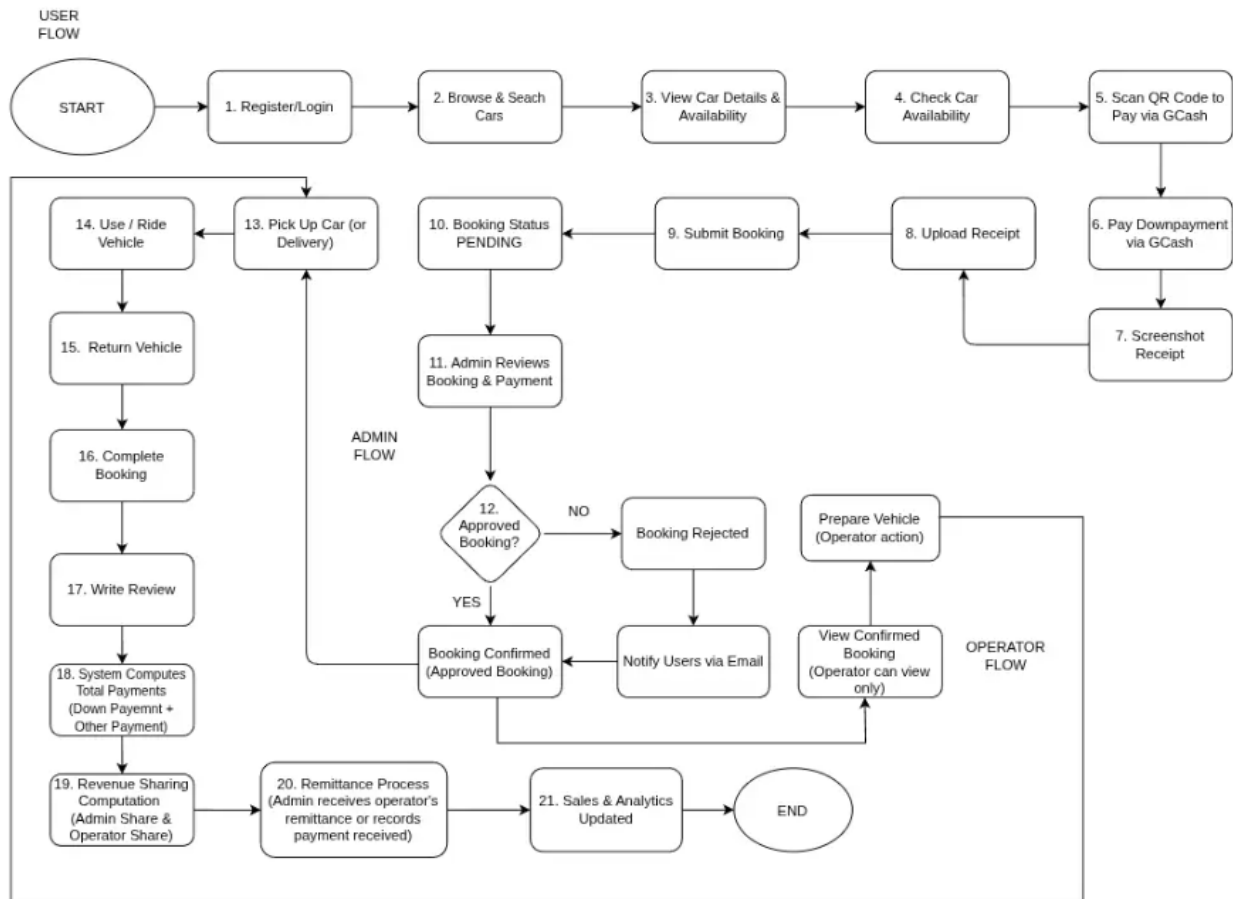
The developed system was created with the following objectives:

- To automate the management of customer, vehicle, and rental records.
- To simplify the reservation and rental transaction process.
- To monitor the availability and current status of rental vehicles.
- To record customer payments and maintain rental histories accurately.
- To generate reports on rentals, revenue, and vehicle utilization.
- To provide analytical dashboards that supported business monitoring and decision-making.
- To improve the overall efficiency, accuracy, and security of car rental operations.

The development of the **Car Rental Management and Analytics System** followed modern software development practices. It incorporated a responsive user interface, secure login and authentication, centralized database management, and reporting tools that made daily operations more organized and efficient. These features ensured that only authorized users could access the system while maintaining the accuracy and security of stored information.

The completed system consisted of several integrated modules that worked together to support the different functions of a car rental business. These modules included user authentication, customer management, vehicle management, reservation processing, rental management, payment management, maintenance monitoring, and analytics reporting. Since all modules shared the same database, information was updated automatically, allowing users to view accurate and real-time records throughout the system.

Figure 4.1 presents the overall workflow of the developed system.



The process began when the user either created a new account or logged into the system using valid credentials. After successfully accessing the system, the user was able to browse the available vehicles, search for a preferred car, and view detailed information such as the vehicle's specifications, rental rate, and availability. Once the user selected a suitable vehicle, the system allowed them to proceed with the booking process.

To confirm the reservation, the user scanned the provided GCash QR code and paid the required down payment. After completing the payment, the user captured a screenshot of the transaction receipt and uploaded it to the system before submitting the booking request. At this stage, the booking status was automatically marked as **Pending**, indicating that it was waiting for verification.

The administrator then reviewed the submitted booking information together with the uploaded payment receipt. If the payment and booking details were valid, the reservation was approved and its status changed to **Booking Confirmed**. If any issues were found during the verification process, the booking was rejected, and the system automatically sent an email notification to inform the user of the decision.

Once the booking had been approved, the operator accessed the confirmed reservation and prepared the assigned vehicle for release. After the vehicle was ready, the user either picked it up at the designated location or received it through the available delivery service. The customer then used the rented vehicle for the agreed rental period before returning it on the scheduled date.

After the vehicle had been returned, the booking was marked as completed. The customer was then given the opportunity to submit a review by providing a rating and feedback regarding their rental experience. The system also automatically calculated the total rental payment by combining the required down payment with any remaining balance. In addition, it computed the revenue sharing between the administrator and the operator based on the completed transaction. The remittance process was then carried out to record the operator's payment or confirm that the required remittance had been received.

Finally, all completed transactions were securely stored in the system's centralized database. The recorded information was automatically reflected in the **Sales and Analytics** module, where it was presented through reports and analytical summaries. These reports enabled the administrator to monitor bookings, payments, revenue, and overall business performance, providing accurate and organized information that supported efficient operations and informed business decision-making.

4.3 System Development Methodology

The development of the **Car Rental Management and Analytics System** adopted the **Agile Software Development Methodology**, an iterative and flexible approach that focused on continuous improvement, collaboration, and regular feedback throughout the software development life cycle. The researchers selected Agile because it provided a structured yet adaptable framework that allowed the project to progress through small, manageable stages. This methodology also enabled the researchers to make necessary improvements whenever new requirements or suggestions were identified during the development process. Instead of developing the entire system at once, the project was completed gradually through several iterations, ensuring that each feature functioned correctly before moving on to the next stage.

Throughout the development process, regular communication with the client and continuous evaluation of the developed modules helped ensure that the system addressed the actual needs of the car rental business. Every completed feature was reviewed, tested, and refined based on observations and feedback before additional functionalities were implemented. This approach minimized development risks, reduced

software errors, and improved the overall quality of the system. By following the Agile methodology, the researchers were able to produce a reliable, user-friendly, and efficient application that met the objectives of the study.

The development of the **Car Rental Management and Analytics System** consisted of the following phases:

Requirements Gathering

The first phase involved gathering the necessary information needed for the development of the system. The researchers conducted interviews with the client and carefully observed the existing workflow of the car rental business to gain a better understanding of its daily operations. Existing documents, transaction records, and business procedures were also reviewed to identify the strengths and weaknesses of the current process.

During this phase, the researchers identified several challenges experienced by the business, including manual recording of rental transactions, difficulty in monitoring vehicle availability, delayed report generation, inconsistent record management, and the lack of analytical tools for monitoring business performance. The information collected from the client became the foundation for identifying both the functional and non-functional requirements of the proposed system. These requirements guided the researchers throughout the entire development process and ensured that the final product addressed the actual operational needs of the business.

System Planning

After completing the requirements gathering phase, the researchers proceeded with planning the project. This phase involved defining the scope and limitations of the study, establishing the objectives of the system, preparing the development timeline, and identifying the software and hardware resources needed for implementation.

The researchers also determined the different modules that would be included in the system, such as user authentication, customer management, vehicle management, reservation processing, rental management, payment recording, maintenance monitoring, analytics reporting, and report generation. A feasibility analysis was also conducted to evaluate the project's technical, operational, and economic feasibility. This planning process ensured that the project remained organized and that all development activities were completed according to schedule.

System Design

During the system design phase, the researchers prepared the overall structure and architecture of the application before the actual coding process began. The system's workflow, database structure, user interface, and functional modules were carefully designed to ensure that each component would operate efficiently and communicate properly with one another.

Several software design tools were utilized during this phase, including **Unified Modeling Language (UML) diagrams**, **Entity Relationship Diagrams (ERDs)**, wireframes, flowcharts, and data flow diagrams. These design documents served as blueprints that guided the researchers during software development.

Special consideration was also given to the user interface to ensure that the application would be simple, organized, and easy to navigate. Menu layouts, forms, dashboards, buttons, and reports were designed with usability in mind so that administrators could perform their daily tasks with minimal difficulty. The database was also carefully designed to organize customer records, vehicle information, rental transactions, payment details, maintenance records, and analytical data while maintaining data accuracy and consistency.

Software Development

After completing the system design, the researchers proceeded with the actual development of the application. During this phase, the planned designs were translated into a fully functional system using the selected programming languages, frameworks, and database management system.

The researchers developed the system incrementally by focusing on one module at a time. Each completed module was integrated into the existing system before additional features were implemented. This development strategy allowed the researchers to detect and correct errors immediately, reducing the possibility of major software issues later in the project.

The developed system included several major functionalities, such as secure user authentication, customer registration, vehicle information management, reservation processing, rental transaction management, payment recording, maintenance scheduling, report generation, and an analytics dashboard. The analytics component summarized rental transactions into graphical reports and statistical summaries that allowed administrators to monitor vehicle utilization, rental trends, and business performance. These features helped transform raw transaction records into useful information that supported better planning and decision-making.

Testing

Testing played an essential role in ensuring the quality, reliability, and functionality of the developed system. Throughout the development process, every completed module underwent **unit testing** to verify that each function performed according to its intended purpose. Errors identified during testing were immediately corrected before the module was integrated into the complete application.

After all modules had been completed, **integration testing** was conducted to verify that the interaction among the different modules functioned correctly. The researchers ensured that information entered into one module was accurately reflected throughout

the entire system. Once integration testing had been completed successfully, **system testing** was performed to evaluate the application's overall performance, security, reliability, and responsiveness under normal operating conditions.

To determine whether the developed application met the expectations of its intended users, **User Acceptance Testing (UAT)** was conducted with selected respondents. During this stage, the respondents evaluated the system based on its functionality, usability, performance, reliability, security, maintainability, and portability using the **ISO/IEC 25010 Software Quality Model**. The comments and suggestions gathered from the respondents were carefully reviewed and used to implement final improvements before deployment.

Deployment

After successfully completing all testing activities, the researchers deployed the **Car Rental Management and Analytics System** in the intended operating environment. The application was installed and configured together with its database, ensuring that all system components functioned properly. User accounts were created for authorized personnel, and the necessary system settings were configured before the application became operational.

The researchers also conducted an orientation for the administrator to demonstrate the proper use of the system. The training included customer registration, vehicle management, reservation processing, rental transactions, payment recording, maintenance monitoring, report generation, and the use of the analytics dashboard. Sample customer records, vehicle information, and rental transactions were entered into the system to verify that all modules worked correctly in an actual business environment. The deployment process confirmed that the application was stable, functional, and ready for daily use.

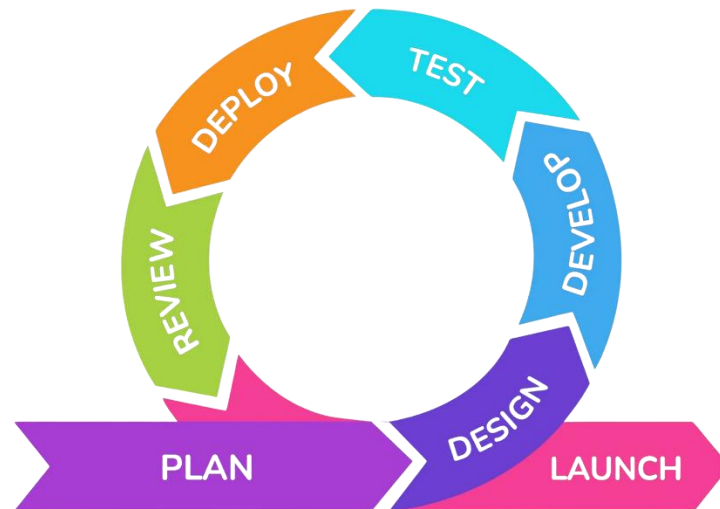
Maintenance

The final phase of the Agile Software Development Methodology involved maintaining and improving the system after deployment. During this phase, the researchers continuously monitored the application's performance and corrected any issues that were discovered during actual operation. Minor software updates were implemented whenever necessary to improve functionality, enhance system security, and optimize overall performance.

Feedback provided by the client and users was also collected and evaluated to identify possible enhancements for future versions of the application. Suggestions such as integrating online payment methods, adding SMS or email notifications, improving the analytics dashboard, and developing a mobile version of the application were considered as possible future improvements. Continuous maintenance ensured that the system remained reliable, secure, and capable of supporting the long-term operational needs of the car rental business.

Figure 4.2 illustrates the **Agile Software Development Life Cycle** adopted in the development of the **Car Rental Management and Analytics System**.

(Insert Agile Software Development Life Cycle Diagram Here)



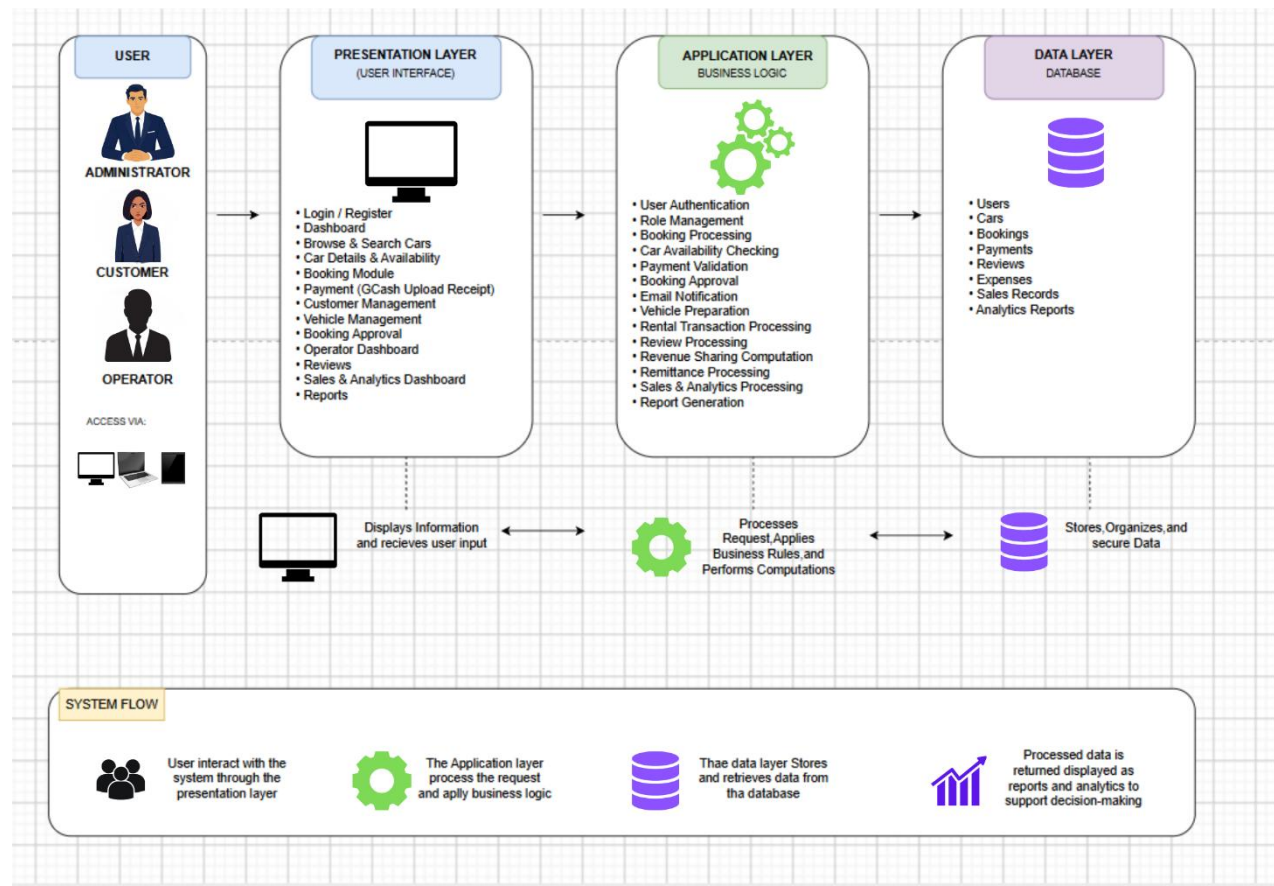
4.4 System Architecture

The **Car Rental Management and Analytics System** utilized a **Three-Tier Client-Server Architecture**, which divided the system into three independent but interconnected layers: the **Presentation Layer**, **Application Layer**, and **Data Layer**. This architectural design was selected because it improved the organization of the system by separating the user interface, business logic, and database management into different layers. As a result, the system became easier to maintain, more secure, and capable of handling future enhancements without significantly affecting the existing functionalities.

The Three-Tier Architecture also allowed the application to process user requests more efficiently by assigning specific responsibilities to each layer. This separation reduced system complexity, improved maintainability, and ensured that data could be securely managed while providing users with a smooth and responsive experience. The architecture supported the overall goal of developing a reliable and scalable system that

could efficiently manage car rental operations and generate analytical reports for business decision-making.

Figure 4.3 illustrates the overall system architecture of the developed **Car Rental Management and Analytics System**.



The architecture consisted of the following components:

Presentation Layer

The **Presentation Layer** served as the front-end interface of the application where administrators and authorized users interacted with the system. It was responsible for displaying information, receiving user input, and presenting reports and analytics in an organized and user-friendly manner. This layer was designed to provide a simple and responsive interface that allowed users to perform tasks efficiently without requiring advanced technical knowledge.

Users were able to access the system through desktop computers, laptops, tablets, or smartphones using a modern web browser connected to the internet or a local network, depending on the deployment setup. The interface was designed with consistency and

ease of navigation in mind, enabling users to quickly locate system functions and complete daily operations with minimal effort.

The major interface components included the following:

- **Login Page** – Allowed authorized users to securely access the system by entering valid login credentials.
- **Dashboard** – Displayed an overview of important business information, including the total number of rentals, available vehicles, rented vehicles, revenue summaries, and other key performance indicators.
- **Customer Management** – Enabled administrators to register, update, search, and manage customer information.
- **Vehicle Management** – Allowed users to add, edit, monitor, and update vehicle details, including availability and maintenance status.
- **Reservation Management** – Provided tools for processing customer reservations and confirming rental schedules.
- **Rental Management** – Recorded rental transactions, monitored ongoing rentals, and managed vehicle returns.
- **Payment Management** – Recorded rental payments, outstanding balances, and payment histories.
- **Maintenance Management** – Allowed administrators to schedule vehicle maintenance and monitor vehicles currently undergoing repair or servicing.
- **Analytics Dashboard** – Displayed graphical reports, charts, and statistical summaries related to rental performance, vehicle utilization, and business operations.
- **Reports** – Generated printable reports for rentals, payments, customers, vehicle status, and overall business performance.

These interface components worked together to provide users with an efficient and organized platform for managing all aspects of the car rental business.

Application Layer

The **Application Layer** contained the core business logic of the system. It processed every request submitted by users through the Presentation Layer and ensured that all business rules were correctly implemented before any information was stored or retrieved from the database. This layer acted as the communication bridge between the user interface and the database, ensuring that data processing was accurate, secure, and efficient.

The Application Layer validated user input, managed system operations, processed rental transactions, and generated analytical reports based on the stored data. It also handled user authentication and ensured that only authorized users could access specific system functions according to their assigned roles.

Its primary responsibilities included:

- User authentication and access control
- Customer information management

- Vehicle registration and management
- Reservation processing
- Rental transaction processing
- Payment recording and monitoring
- Vehicle availability monitoring
- Maintenance scheduling and status updates
- Revenue computation
- Rental trend analysis
- Vehicle utilization analysis
- Dashboard generation
- Report generation

Whenever a user performed an action, such as adding a new customer, processing a reservation, recording a payment, or updating vehicle information, the Application Layer validated the request before communicating with the Data Layer. This process ensured that only accurate and valid information was stored in the database, reducing errors and maintaining data consistency throughout the system.

The analytics component was also processed within this layer. Rental records and payment transactions were analyzed to generate useful business insights, such as the most rented vehicles, monthly rental trends, revenue summaries, vehicle utilization rates, and other statistical information that supported managerial decision-making.

Data Layer

The **Data Layer** served as the central repository of all information used by the **Car Rental Management and Analytics System**. It was responsible for storing, organizing, updating, and retrieving data whenever requested by the Application Layer. A relational database management system was used to maintain data consistency, reduce redundancy, and ensure the integrity of all stored information.

The database stored all operational and analytical data required for the daily activities of the business. Proper relationships between database tables were established using primary keys and foreign keys to ensure that information remained accurate and properly linked across different modules.

The stored information included:

- User Accounts
- Customer Information
- Vehicle Records
- Vehicle Categories
- Reservations
- Rental Transactions
- Payment Records
- Maintenance Records
- Vehicle Availability Status
- Revenue Records

- Analytics Data
- Generated Reports

The centralized database allowed information entered in one module to become immediately available to other related modules. For example, once a rental transaction was completed, the system automatically updated the vehicle's availability, recorded the payment details, and reflected the transaction in the analytics dashboard and reports. This eliminated duplicate encoding and improved the accuracy and reliability of the stored information.

The Data Layer also implemented security mechanisms to protect sensitive information from unauthorized access. User credentials were securely stored, while access permissions ensured that only authorized personnel could modify or retrieve confidential business records. Regular database management practices further helped maintain data integrity, system reliability, and efficient retrieval of information.

Overall, the Three-Tier Client-Server Architecture provided a strong foundation for the **Car Rental Management and Analytics System**. By separating the presentation, application, and data layers, the system became easier to manage, more secure, and capable of supporting future enhancements. This architecture also ensured smooth communication between users, the application, and the database, resulting in a reliable system that effectively supported the daily operations and decision-making needs of the car rental business.

4.5 Hardware and Software Requirements

The successful development, implementation, and operation of the **Car Rental Management and Analytics System** required appropriate hardware and software resources. These resources played an important role in ensuring that the system functioned efficiently during development, testing, deployment, and actual use. Selecting suitable hardware and software specifications helped provide a stable environment where the application could process rental transactions, manage customer and vehicle records, generate analytical reports, and store business information without significant delays or performance issues.

During the development phase, the researchers utilized hardware with sufficient processing power and memory to support software development tools, database management, and system testing. Reliable hardware resources allowed the researchers to run multiple applications simultaneously, such as the code editor, local server, database management system, and web browser, without affecting the overall performance of the system.

Similarly, appropriate software tools were selected based on their compatibility, reliability, ease of use, and widespread adoption in modern web application development. These software applications supported the design, coding, debugging, testing, and deployment of the **Car Rental Management and Analytics System**. The selected technologies also provided flexibility for future maintenance and possible system enhancements.

The recommended hardware and software specifications ensured that the application could efficiently manage rental transactions, monitor vehicle availability, generate business reports, and perform analytics while maintaining system security, data accuracy, and responsiveness. These requirements also ensured that authorized users could operate the system smoothly under normal business conditions.

4.5.1 Hardware Requirements

Table 4.1 presents the recommended hardware specifications.

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	256 GB SSD	512 GB SSD
Monitor	1366 × 768 Resolution	Full HD Display
Internet Connection	10 Mbps	50 Mbps
Mobile Device	Android 10	Android 13 or higher

The recommended hardware configuration provides sufficient processing capability for executing analytical operations and rendering graphical dashboards without significant delays.

4.5.2 Software Requirements

Table 4.2 presents the software requirements.

Software	Purpose
Windows 11 / Ubuntu Linux	Operating System

Visual Studio Code	Source Code Editor
MySQL	Database Management System
PHP / Node.js	Backend Development
ReactJS	Frontend Development
Google Chrome	Web Browser
Git	Version Control
XAMPP	Local Development Server

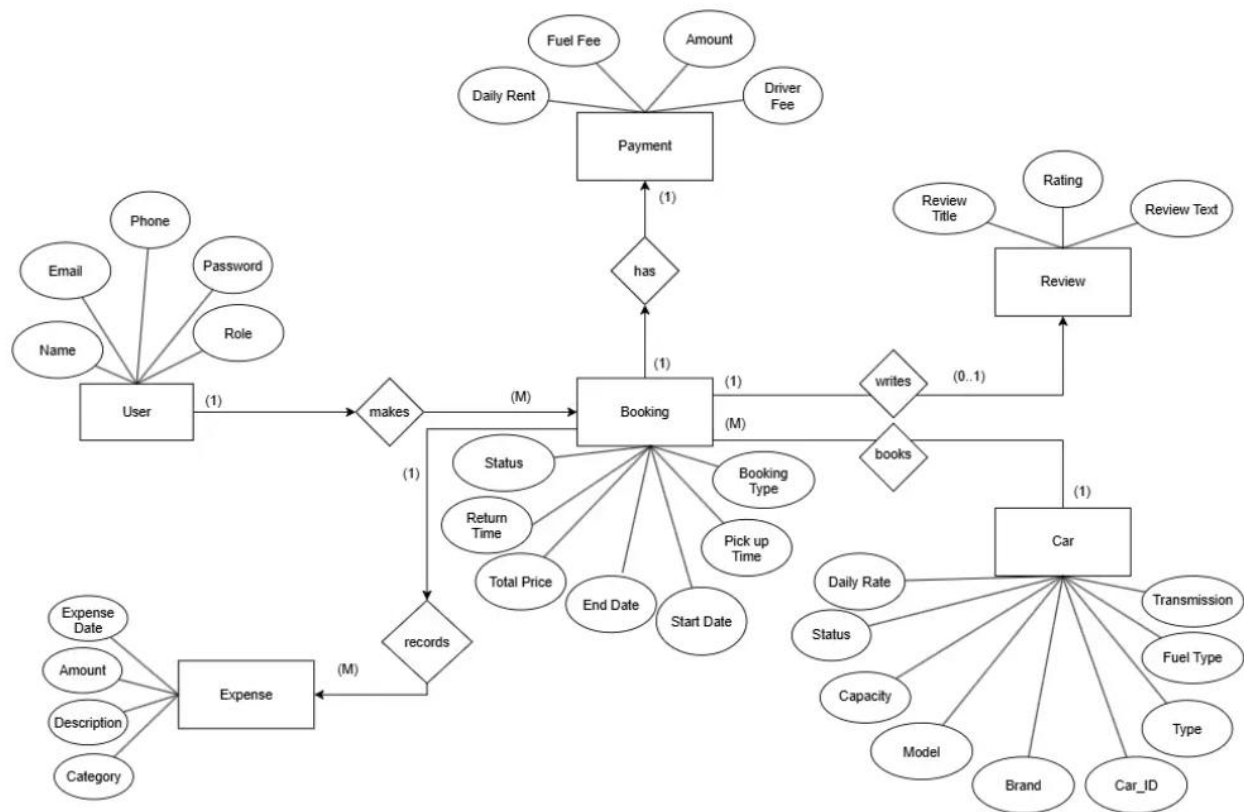
The selection of these software tools was based on their stability, community support, compatibility, and widespread adoption in modern web application development.

4.6 Database Design

The database serves as the foundation of the TOPBuy App by organizing and storing transaction records, customer information, product inventories, and analytical results. A well-designed database ensures efficient data retrieval, minimizes redundancy, and maintains data integrity throughout the operation of the system.

The researchers employed a **relational database model** using MySQL. Database normalization techniques were applied to reduce duplication and maintain consistency among related tables.

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.



The database of the **Car Rental Management and Analytics System** was designed to provide a centralized and well-organized repository for all information required by the application. It stored user accounts, vehicle records, booking transactions, payment information, customer reviews, and business expenses. The database was structured using a relational database model to ensure data consistency, minimize redundancy, and maintain the integrity of information throughout the system.

Each table was assigned a primary key to uniquely identify every record, while foreign keys were used to establish relationships among related tables. These relationships allowed the system to retrieve and process data efficiently while maintaining referential integrity. The organized database structure also supported the generation of reports and analytical summaries used by administrators in monitoring business operations.

The primary entities of the database included the following:

- Users
 - Cars
 - Bookings
 - Payments
 - Reviews
 - Expenses
-

Users Table

The **Users** table stored the account credentials and profile information of authorized users who could access the system. It was responsible for authentication and determining the access privileges of each user.

Field	Description
user_id	Primary Key
name	Full Name of the User
email	User Email Address
phone	Contact Number
password	Encrypted Password
role	User Role (Administrator or Staff)

Cars Table

The **Cars** table stored detailed information about every vehicle available for rent. It also maintained the current status of each vehicle to help administrators monitor availability.

Field	Description
car_id	Primary Key
brand	Vehicle Brand

Field	Description
model	Vehicle Model
type	Vehicle Type
transmission	Transmission Type
fuel_type	Fuel Type
capacity	Seating Capacity
daily_rate	Daily Rental Rate
status	Vehicle Status

Bookings Table

The **Bookings** table recorded all reservation and rental transactions made by customers. It served as the central table of the system because it connected users, vehicles, payments, and reviews.

Field	Description
booking_id	Primary Key
user_id	Foreign Key
car_id	Foreign Key
start_date	Rental Start Date
end_date	Rental End Date
pickup_time	Vehicle Pick-up Time
return_time	Vehicle Return Time
booking_type	Booking Type
total_price	Total Rental Cost
status	Booking Status

Payments Table

The **Payments** table stored all payment information associated with rental transactions. It recorded the different charges incurred during the rental period and the total amount paid by the customer.

Field	Description
payment_id	Primary Key
booking_id	Foreign Key
daily_rent	Daily Rental Fee
driver_fee	Driver Service Fee
fuel_fee	Fuel Charge
amount	Total Payment Amount

Reviews Table

The **Reviews** table stored customer feedback submitted after completing a rental transaction. These records allowed administrators to evaluate customer satisfaction and improve service quality.

Field	Description
review_id	Primary Key
booking_id	Foreign Key
review_title	Review Title
review_text	Customer Feedback

Field	Description
rating	Customer Rating

Expenses Table

The **Expenses** table recorded the operational expenses of the car rental business. It helped administrators monitor business expenditures and maintain financial records.

Field	Description
expense_id	Primary Key
category	Expense Category
description	Expense Description
amount	Expense Amount
expense_date	Date of Expense

The relationships established among these tables enabled the system to efficiently manage rental operations while maintaining accurate and consistent records. The **Users** table was linked to the **Bookings** table, allowing the system to identify the user responsible for each booking transaction. Each booking was associated with a specific vehicle through the **Cars** table and was connected to its corresponding **Payment** record to ensure accurate tracking of rental charges. Completed bookings could also be linked to customer **Reviews**, providing valuable feedback regarding the quality of service. In addition, the **Expenses** table maintained records of operational costs, allowing administrators to monitor business expenditures alongside rental income.

CHAPTER 5

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CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter concluded the study by presenting the overall findings, conclusions, and recommendations derived from the development and evaluation of the **Car Rental Management and Analytics System**. It discussed whether the developed system successfully fulfilled the objectives of the research and addressed the challenges encountered in the existing car rental management process. The results obtained during system testing and evaluation were carefully analyzed to determine the effectiveness and overall performance of the application.

In addition, this chapter summarized the significant contributions of the developed system in improving the management of vehicle rentals, customer records, reservations, payments, and business reporting. It also highlighted how the analytics features supported business owners by providing useful insights into rental activities and operational performance, allowing them to make more informed decisions based on accurate and organized information.

Finally, this chapter presented recommendations for future improvements that could further enhance the functionality, usability, and reliability of the system. These recommendations served as valuable references for future researchers and developers who may continue or expand the study by integrating additional technologies and

advanced features. Through these discussions, the chapter emphasized the importance of adopting modern information systems to improve efficiency, productivity, and decision-making in car rental businesses.

5.2 Summary of Findings

The development and evaluation of the **Car Rental Management and Analytics System** demonstrated that the objectives of the study were successfully achieved. The findings showed that the developed system effectively addressed the challenges associated with the manual management of car rental operations by providing a centralized and automated platform for managing business transactions.

Based on the results of the study, the following findings were obtained:

- The developed system successfully automated the management of customer information, vehicle records, reservations, rental transactions, payment processing, and maintenance monitoring, resulting in faster and more organized business operations.
- The system effectively monitored the availability and status of rental vehicles, allowing administrators to easily identify which vehicles were available, rented, or undergoing maintenance. This improved the accuracy of vehicle tracking and reduced scheduling conflicts.
- The analytics dashboard successfully transformed rental transaction data into meaningful reports and graphical summaries. These reports enabled business owners to monitor rental trends, vehicle utilization, and revenue performance, providing valuable information that supported better planning and business decision-making.
- The centralized database improved the accuracy, consistency, and security of business records by minimizing duplicate entries, reducing manual errors, and allowing authorized users to retrieve information quickly whenever needed.
- The system generated comprehensive reports that simplified the monitoring of daily rental activities, customer transactions, payments, and overall business performance. These reports reduced the time required to prepare manual records and supported more efficient business management.
- Based on the results of the system evaluation conducted using the **ISO/IEC 25010 Software Quality Model**, the respondents found the system to be functional, reliable, user-friendly, efficient, secure, maintainable, and portable. The evaluation results indicated that the developed application met the quality standards expected for a modern web-based information system.
- The User Acceptance Testing (UAT) further confirmed that the system was easy to understand and operate. The respondents were able to perform the required tasks with minimal difficulty, demonstrating that the application's interface and functionalities were appropriate for the intended users.

Overall, the findings of the study showed that the **Car Rental Management and Analytics System** successfully improved the efficiency of car rental operations by automating essential business processes and providing analytical tools that supported informed decision-making. The system effectively met the objectives of the study and demonstrated its potential to improve the overall management and performance of car rental businesses.

5.3 Conclusion

The **Car Rental Management and Analytics System** proved to be an effective and practical solution for improving the daily operations of car rental businesses. By automating essential processes such as customer management, vehicle monitoring, reservation handling, rental transactions, payment recording, and report generation, the system reduced the limitations associated with manual record-keeping and improved the overall efficiency of business operations.

The integration of analytics features enabled the system to transform rental transaction data into meaningful reports and visual summaries. These analytical tools provided business owners with valuable insights into rental trends, vehicle utilization, and revenue performance, allowing them to make more informed decisions regarding business operations and resource management. Instead of relying solely on manual records, administrators were able to monitor the performance of the business using accurate and organized information generated by the system.

The evaluation results also demonstrated that the developed system successfully met the objectives of the study. Based on the respondents' assessment using the **ISO/IEC 25010 Software Quality Model**, the system performed well in terms of functionality, usability, reliability, performance efficiency, security, maintainability, and portability. These findings indicated that the application was capable of meeting the operational requirements of a car rental business while providing a user-friendly and dependable platform for its intended users.

Overall, the study concluded that the **Car Rental Management and Analytics System** served as a reliable and efficient solution for modernizing car rental operations. The developed system not only simplified business processes but also supported data-driven decision-making through its reporting and analytics capabilities. Therefore, the researchers concluded that the system successfully achieved the objectives of the study and demonstrated its potential to enhance the productivity, accuracy, and overall performance of car rental businesses.

5.4 Recommendations

Based on the findings and conclusions of the study, the researchers proposed the following recommendations to further improve the **Car Rental Management and Analytics System** and enhance its effectiveness for future implementation:

System Enhancements

- The system could be enhanced by integrating **online payment gateways** to allow customers to make secure and convenient online payments using digital wallets, bank transfers, or credit and debit cards.
- An **SMS and email notification feature** could be added to automatically remind customers about reservation confirmations, payment due dates, rental schedules, vehicle return dates, and maintenance updates. This feature would improve communication between the business and its customers.
- Future versions of the system could include a **mobile application** for Android and iOS devices to provide users with greater accessibility and allow administrators to monitor rental activities anytime and anywhere.
- The analytics dashboard could be expanded by incorporating **advanced reporting and forecasting features**, such as predicting peak rental seasons, identifying customer rental patterns, and estimating future business demand. These additional insights could further support strategic planning and business decision-making.
- A **GPS vehicle tracking feature** could also be integrated to allow administrators to monitor the real-time location of rented vehicles, improving vehicle security and operational efficiency.

Recommendations for Future Researchers

- Future researchers may consider integrating **artificial intelligence (AI)** or **machine learning** techniques to improve the analytics capabilities of the system. These technologies could be used to predict customer demand, recommend pricing strategies, and analyze rental behavior more accurately.
- Similar studies may also be conducted using a larger number of respondents or different car rental businesses to further evaluate the system's performance, usability, and scalability in various operational environments.
- Future researchers are encouraged to enhance the security features of the system by implementing advanced authentication methods, stronger data encryption, and additional cybersecurity measures to better protect sensitive customer and business information.

Implementation Recommendations

- Business owners who intend to adopt the system should provide proper training to administrators and staff to ensure that all system features are utilized effectively and consistently during daily operations.
- Regular database backup and preventive system maintenance should be performed to protect business records, maintain data integrity, and minimize the risk of data loss or system failure.
- Finally, businesses should periodically review and update the system based on user feedback and changing operational requirements. Continuous improvements would help

ensure that the application remained reliable, efficient, and capable of supporting the long-term needs of the car rental business.

Overall, these recommendations were proposed to support the continuous improvement of the **Car Rental Management and Analytics System** and to encourage future researchers and developers to further expand its capabilities. By implementing these suggestions, the system could become more adaptable to technological advancements and better equipped to meet the evolving needs of the car rental industry.

System Improvements

- Integrate automated notification alerts for low-stock items and expiring products to further improve inventory management.
- Enhance security protocols by implementing two-factor authentication to better protect sensitive sales and customer data.
- Develop advanced export features allowing reports to be generated in multiple formats (e.g., PDF, CSV, Excel) for easier sharing and integration with accounting software.

Future Research Directions

- Explore the integration of machine learning algorithms to enable predictive analytics, such as forecasting future sales trends based on seasonal data.
- Conduct a longitudinal study to analyze the long-term impact of the system on business growth and profitability for SMEs.
- Investigate the feasibility of a mobile-native application to provide offline functionality for businesses with intermittent internet connectivity.

Implementation Strategies

- Business owners should conduct training sessions for staff to ensure consistent data entry habits, which are critical for accurate analytics.
- SMEs are encouraged to perform regular data backups and system maintenance to ensure security and prevent data loss.

5.5 Impact of the Study

The development of the **Car Rental Management and Analytics System** made a meaningful contribution to the field of Information Technology by demonstrating how a web-based management system combined with data analytics could improve the operations of a car rental business. The study showed that integrating information management and analytical tools into a single platform could help automate business processes, reduce manual work, and provide reliable information for decision-making. This highlighted the importance of using technology to address common operational challenges faced by small and medium-sized enterprises.

For the business owner, the developed system provided a more organized and efficient way of managing customer records, vehicle information, reservations, rental transactions, payments, and maintenance schedules. The analytics feature further enhanced the value of the system by presenting rental trends, vehicle utilization, and revenue reports in an easy-to-understand format. These insights supported better planning, resource allocation, and overall business management.

The study also benefited future researchers and software developers by serving as a reference for developing similar information systems in the transportation and rental service industries. The methodologies, design approaches, and evaluation process presented in this research could guide future studies that aim to improve or expand existing rental management systems through the integration of emerging technologies and advanced analytical features.

Although the developed system successfully achieved its intended objectives, the researchers recognized that there were opportunities for further improvement. Additional features such as online payment integration, GPS vehicle tracking, mobile application support, and predictive analytics could further increase the system's functionality and effectiveness. Despite these limitations, the findings of the study demonstrated that adopting a computerized management system with analytics capabilities could significantly improve operational efficiency, data accuracy, and decision-making in car rental businesses.

Overall, the study emphasized the growing role of information technology in modern business operations. It demonstrated that even small and medium-sized car rental businesses could benefit from digital transformation by adopting systems that not only automated routine tasks but also generated valuable business insights. The successful implementation of the **Car Rental Management and Analytics System** highlighted the potential of technology to enhance productivity, improve customer service, and support sustainable business growth.

5.6 Summary

In summary, the **Car Rental Management and Analytics System** successfully demonstrated how information technology could be utilized to improve the efficiency and effectiveness of car rental operations. Through the integration of automated management functions and analytical reporting tools, the developed system addressed the limitations of manual processes and provided a more organized, accurate, and reliable way of managing business transactions.

The study achieved its objectives by developing a web-based system that streamlined customer management, vehicle monitoring, reservation processing, rental transactions, payment recording, maintenance tracking, and report generation. The inclusion of an analytics dashboard further enhanced the system by providing valuable insights into rental trends, vehicle utilization, and business performance, enabling business owners to make informed and timely decisions.

The evaluation results confirmed that the developed system met the expected software quality standards and was positively accepted by the selected respondents. The findings demonstrated that the system was functional, user-friendly, reliable, secure, and efficient in supporting the daily operations of a car rental business.

Overall, this research provided a practical solution to the challenges faced by car rental businesses while contributing to the continuous adoption of digital technologies in business management. It also established a strong foundation for future studies and system enhancements that could further improve the capabilities of the application through the integration of emerging technologies and additional features. The researchers concluded that the **Car Rental Management and Analytics System** has the potential to become a valuable tool for improving operational efficiency, enhancing customer service, and supporting sustainable business growth in the car rental industry.